**STA 475 Final Project - Ice Cover Project - Team 1**

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**Formal Analysis**

## Research Question

Looking at the daily data for ice coverage in the Great Lakes, is there any trend or pattern that shows a change in the amount of ice in the lakes? We are interested in aggregating the data for each year, to explore the change between different seasons. As a result of climate change, there is a concern that there is periodically less ice on Earth. This possibility will be explored on the data from the Great Lakes, investigating if there is a significant decrease in the amount of ice.

## Data Manipulation

The existence of NAs on a lot of days raises multiple questions that will affect how we treat the data. First, the starting day for the measurements, and the magnitude of this first measurement seem to vary significantly from year to year, especially comparing the early years to later data. In these early years, the first measurement can be very high, which seems unlikely that such a rise in ice coverage happened from one day to the next. This could mean that in early years, the data started being recorded later, or a less accurate measurement instrument that may not detect lower percentages of ice as well as modern instruments. Therefore, an interpolation of data will be necessary in these years when the first measurement is very high. Another important change will be aggregating the daily data into one year to reflect the amount of ice the lakes had in one year, so that a year to year comparison can be made. We explored three aggregation methods: Area Under Curve (Sum of Ice Coverage), Average Ice Coverage, and Maximum Ice Coverage.

**Analysis**

Our team was tasked with approaching this using parametric methods. We initially wanted to use a Simple Linear Regression Model with our aggregated ice coverage statistic being predicted by Year. However, we analyzed the data and found that for all aggregation methods the statistic at one year was negatively correlated with the statistic of the previous year (one lag). So to account for the effect of one lag on the outcome we fit a Multiple Linear Regression model including the previous year's statistic as a predictor also..

**Challenges**

Violation of the independence assumption makes linear regression more complex. We must account for the correlation between years when using the year as a predictive factor. Wide variation throughout the seasons, and short-term changes in year to year ice coverage makes visuals noisy and overall interpretation difficult.

**Results**

From all our models, there seems to be an overall decreasing trend in ice coverage from 1973 to recent years.